QUASI-COHERENT MODULES ARE FILTERED COLIMITS OF FINITELY PRESENTED ONES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2498-2512, following the statement at 2491-2496.

Editorial note. No omissions; the argument is a genuine observation plus a named-engine consumption.

Allowed standard prerequisites (editorial).

- Noetherian approximation of qcqs spaces. proposition-approximate (published tag 07SU, cited at ~2501-2502). X = lim X_i with X_i of finite presentation over Spec(Z).
- fp implies Noetherian; affine transitions. Morphisms of Spaces, Lemma spaces-morphisms-lemma-finite-presentation-noetherian (cross-chapter, cited at ~2503-2505); lemma-directed-inverse-system-has-limit (same chapter, card L26291, cited at ~2506-2507). X_i is Noetherian; X -> X_i is affine.
- Colimits of qc modules on Noetherian spaces. Cohomology of Spaces, Lemma spaces-cohomology-lemma-direct-colimit-finite-presentation (cross-chapter, cited at ~2508-2511). Concluding engine for affine morphisms to Noetherian spaces.

Source reference. Original lemma 07SF: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 878–878. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 07SU: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2328–2328. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07UW: focused-seeds/geometry/S9-spaces-limits-colimit-finitely-presented/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03I5: focused-seeds/geometry/S9-spaces-limits-colimit-finitely-presented/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction

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Source reference. Original reference 04ZL: focused-seeds/geometry/S9-spaces-limits-colimit-finitely-presented/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03BS: focused-seeds/geometry/S9-spaces-limits-colimit-finitely-presented/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S. Every quasi-coherent $\mathcal{O}_X$-module is a filtered colimit of finitely presented $\mathcal{O}_X$-modules.*

Proof. We may view X as an algebraic space over $\mathrm{Spec}(\mathbf{Z})$, see Spaces, Definition 03I5 and Properties of Spaces, Definition 03BS. Thus we may apply Proposition 07SU and write $X = \lim X_i$ with X_i of finite presentation over $\mathbf{Z}$. Thus X_i is a Noetherian algebraic space, see Morphisms of Spaces, Lemma 04ZL. The morphism $X \to X_i$ is affine, see Lemma 07SF. Conclusion by Cohomology of Spaces, Lemma 07UW. □